smacc2::ISmaccComponent
# stateMachine_
# owner_
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()

cl_generic_sensor::
components::CpMessageTimeout
< TMessageType >
+ onMessageTimeout_
+ timeout_
+ postTimeoutMessageEvent
- timeoutTimer_
- initialized_
+ CpMessageTimeout()
+ ~CpMessageTimeout()
+ onMessageTimeout()
+ onComponentInitialization()
+ onInitialize()
# resetTimer()
- timeoutCallback()

smacc2::client_core
_components::CpActionClient
< nav2_msgs::action::NavigateToPose >
+ actionServerName
+ serverTimeout
+ onActionSucceeded_
+ onActionAborted_
+ onActionCancelled_
+ onActionFeedback_
+ onGoalAccepted_
+ onGoalRejected_
+ postSuccessEvent
+ postAbortedEvent
+ postCancelledEvent
+ postFeedbackEvent
- client_
- lastRequest_
- lastCancelResponse_
- feedbackCallback_
- actionMutex_
+ CpActionClient()
+ CpActionClient()
+ ~CpActionClient()
+ sendGoal()
+ cancelGoal()
+ isServerReady()
+ waitForServer()
+ onInitialize()
+ onComponentInitialization()
+ onSucceeded()
+ onAborted()
+ onCancelled()
+ onFeedback()
+ getActionClient()
- onFeedback()
- onResult()
- postResultEvent()

smacc2::client_core
_components::CpActionClient
< ActionT >
+ actionServerName
+ serverTimeout
+ onActionSucceeded_
+ onActionAborted_
+ onActionCancelled_
+ onActionFeedback_
+ onGoalAccepted_
+ onGoalRejected_
+ postSuccessEvent
+ postAbortedEvent
+ postCancelledEvent
+ postFeedbackEvent
- client_
- lastRequest_
- lastCancelResponse_
- feedbackCallback_
- actionMutex_
+ CpActionClient()
+ CpActionClient()
+ ~CpActionClient()
+ sendGoal()
+ cancelGoal()
+ isServerReady()
+ waitForServer()
+ onInitialize()
+ onComponentInitialization()
+ onSucceeded()
+ onAborted()
+ onCancelled()
+ onFeedback()
+ getActionClient()
- onFeedback()
- onResult()
- postResultEvent()

smacc2::client_core
_components::CpTopicSubscriber
< vision_msgs::msg::Detection3DArray >
+ queueSize
+ postMessageEvent
+ postInitialMessageEvent
+ onFirstMessageReceived_
+ onMessageReceived_
- sub_
- firstMessage_
- initialized_
- topicName_
+ CpTopicSubscriber()
+ CpTopicSubscriber()
+ ~CpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ onInitialize()
- messageCallback()

smacc2::client_core
_components::CpTopicSubscriber
< TMessageType >
+ queueSize
+ postMessageEvent
+ postInitialMessageEvent
+ onFirstMessageReceived_
+ onMessageReceived_
- sub_
- firstMessage_
- initialized_
- topicName_
+ CpTopicSubscriber()
+ CpTopicSubscriber()
+ ~CpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ onInitialize()
- messageCallback()

smacc2::client_core
_components::CpTopicSubscriber
< AprilTagDetectionArray >
+ queueSize
+ postMessageEvent
+ postInitialMessageEvent
+ onFirstMessageReceived_
+ onMessageReceived_
- sub_
- firstMessage_
- initialized_
- topicName_
+ CpTopicSubscriber()
+ CpTopicSubscriber()
+ ~CpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ onInitialize()
- messageCallback()

cl_foundation_pose
::CpObjectTracker1
- fondationPoseTopic_
- detectedObjects
+ onInitialize()
+ onDetection3DArrayReceived()
+ getPose()

cl_foundation_pose
::CpObjectTrackerTf
- tfBuffer_
- tfListener_
- tfBroadcaster_
- global_frame_id_
- enabled_
- detectedObjects
- m_mutex_
+ CpObjectTrackerTf()
+ setEnabled()
+ isEnabled()
+ resetPoseEstimation()
+ update()
+ onInitialize()
+ updateAndGetGlobalPose()

cl_gcalcli::CpCalendarEvent
Listener
+ onEventDetected_
+ onEventStarted_
+ onEventEnded_
+ postEventDetectedEvent_
+ postEventStartedEvent_
+ postEventEndedEvent_
- poller_
- watches_
- triggered_events_
- current_events_
- initialized_
- mutex_
+ CpCalendarEventListener()
+ ~CpCalendarEventListener()
+ onInitialize()
+ addWatch()
+ clearWatches()
+ getWatches()
+ resetTriggeredEvents()
+ onEventDetected()
+ onEventStarted()
+ onEventEnded()
+ onStateOrthogonalAllocation()
# update()
- onAgendaUpdated()
- matchesPattern()
- shouldTriggerStart()
- shouldTriggerEnd()
- getTriggeredKey()

cl_gcalcli::CpCalendarPoller
+ onAgendaUpdated_
+ postAgendaUpdatedEvent_
- connection_
- cached_events_
- last_poll_time_
- last_poll_attempt_
- initialized_
- events_mutex_
+ CpCalendarPoller()
+ ~CpCalendarPoller()
+ onInitialize()
+ refreshAgenda()
+ getEvents()
+ findEvents()
+ getEventsInWindow()
+ getActiveEvents()
+ getLastPollTime()
+ onAgendaUpdated()
+ onStateOrthogonalAllocation()
# update()
- parseTsvOutput()
- parseTsvLine()
- parseDateTime()
- generateEventId()

cl_gcalcli::CpGcalcliConnection
+ onConnectionLost_
+ onConnectionRestored_
+ onAuthenticationRequired_
+ postConnectionLostEvent_
+ postConnectionRestoredEvent_
+ postAuthenticationRequired
Event_
- config_
- connection_state_
- consecutive_failures_
- initialized_
- last_heartbeat_time_
- subprocess_executor_
- state_mutex_
+ CpGcalcliConnection()
+ ~CpGcalcliConnection()
+ onInitialize()
+ configure()
+ getConnectionState()
+ isConnected()
+ checkConnection()
+ restartConnection()
+ executeGcalcli()
+ getConfig()
+ onConnectionLost()
+ onConnectionRestored()
+ onAuthenticationRequired()
+ onStateOrthogonalAllocation()
# update()
- performHeartbeat()
- handleConnectionStateChange()
- isAuthenticationError()

cl_generic_sensor::
components::CpMessageTimeout
< MessageType >
+ onMessageTimeout_
+ timeout_
+ postTimeoutMessageEvent
- timeoutTimer_
- initialized_
+ CpMessageTimeout()
+ ~CpMessageTimeout()
+ onMessageTimeout()
+ onComponentInitialization()
+ onInitialize()
# resetTimer()
- timeoutCallback()

cl_http::CpHttpConnection
Manager
- io_context_
- worker_guard_
- io_thread_
- initialized_
+ CpHttpConnectionManager()
+ ~CpHttpConnectionManager()
+ onInitialize()
+ getStrand()
+ getIoContext()

cl_http::CpHttpRequestExecutor
+ onResponseReceived_
- connectionManager_
- sessionManager_
- responseHandler_
- HTTP_VERSION
+ CpHttpRequestExecutor()
+ ~CpHttpRequestExecutor()
+ onInitialize()
+ setDependencies()
+ executeRequest()

cl_http::CpHttpSessionManager
- server_config_
- ssl_context_
- initialized_
+ CpHttpSessionManager()
+ ~CpHttpSessionManager()
+ onInitialize()
+ setServerUrl()
+ createSession()
+ isSSL()
+ getServerName()
+ getPort()

cl_isaac_apriltag::
CpAprilTagMissionState
- mutex_
- visitedWorkingAreas_
- selectedVisitTagId_
+ CpAprilTagMissionState()
+ ~CpAprilTagMissionState()
+ onInitialize()
+ isTagVisited()
+ markTagVisited()
+ clearVisitedTags()
+ getVisitedTags()
+ selectTag()
+ getSelectedTag()
+ clearSelectedTag()

cl_isaac_apriltag::
CpAprilTagTracker
+ onAprilTagDetection_
- topicSubscriber_
- tfBuffer_
- tfListener_
- targetFrame_
- detectedAprilTagsMapPose_
- mutex_
+ CpAprilTagTracker()
+ ~CpAprilTagTracker()
+ onInitialize()
+ onComponentInitialization()
+ getTagsWithinTime()
+ getTagPose()
+ onAprilTagDetected()
- onAprilTagMessage()

cl_isaac_apriltag::
CpAprilTagVisualization
# m_mutex_
# tracker_
# markerPublisher_
# globalApriltagTransform_
+ CpAprilTagVisualization()
+ ~CpAprilTagVisualization()
+ onInitialize()
+ getGlobalApriltagTransform()
+ computeAggregatedMarker()
+ update()

cl_keyboard::components
::CpKeyboardListener1
+ OnKeyPress_
- postEventKeyPress
+ CpKeyboardListener1()
+ ~CpKeyboardListener1()
+ onComponentInitialization()
+ OnKeyPress()
+ onKeyboardMessage()
+ postKeyEvent()

cl_lifecycle_node::
CpLifecycleEventMonitor
+ onTransitionCreate_
+ onTransitionConfigure_
+ onTransitionActivate_
+ onTransitionDeactivate_
+ onTransitionCleanup_
+ onTransitionUnconfigured
Shutdown_
+ onTransitionInactiveShutdown_
+ onTransitionActiveShutdown_
+ onTransitionDestroy_
+ onTransitionOnConfigureSuccess_
+ onTransitionOnConfigureFailure_
+ onTransitionOnConfigureError_
+ onTransitionOnActivateSuccess_
+ onTransitionOnActivateFailure_
+ onTransitionOnActivateError_
+ onTransitionOnDeactivate
Success_
+ onTransitionOnDeactivate
Failure_
+ onTransitionOnDeactivate
Error_
+ onTransitionOnCleanupSuccess_
+ onTransitionOnCleanupFailure_
+ onTransitionOnCleanupError_
+ onTransitionOnShutdownSuccess_
+ onTransitionOnShutdownFailure_
+ onTransitionOnShutdownError_
+ onTransitionOnActiveShutdown
Success_
+ onTransitionOnActiveShutdown
Failure_
+ onTransitionOnActiveShutdown
Error_
+ onTransitionOnErrorSuccess_
+ onTransitionOnErrorFailure_
+ onTransitionOnErrorError_
- nodeName_
- subscription_
- lastTransitionEvent_
- eventMutex_
- postEventConfigure_
- postEventActivate_
- postEventDeactivate_
- postEventCleanup_
- postEventUnconfiguredShutdown_
- postEventInactiveShutdown_
- postEventActiveShutdown_
- postEventDestroy_
- postEventOnConfigureSuccess_
- postEventOnConfigureFailure_
- postEventOnConfigureError_
- postEventOnActivateSuccess_
- postEventOnActivateFailure_
- postEventOnActivateError_
- postEventOnDeactivateSuccess_
- postEventOnDeactivateFailure_
- postEventOnDeactivateError_
- postEventOnCleanupSuccess_
- postEventOnCleanupFailure_
- postEventOnCleanupError_
- postEventOnShutdownSuccess_
- postEventOnShutdownFailure_
- postEventOnShutdownError_
- postEventOnErrorSuccess_
- postEventOnErrorFailure_
+ CpLifecycleEventMonitor()
+ ~CpLifecycleEventMonitor()
+ setOwner()
+ getLastTransitionEvent()
+ onInitialize()
+ onStateOrthogonalAllocation()
- onTransitionEvent()

cl_lifecycle_node::
CpLifecycleStateTracker
- currentState_
- stateMutex_
+ CpLifecycleStateTracker()
+ ~CpLifecycleStateTracker()
+ updateState()
+ getCurrentState()
+ getCurrentStateLabel()
+ getCurrentStateId()
+ onInitialize()

cl_mission_tracker
::CpDecisionManager
- mutex_
- decisionCounter_
+ CpDecisionManager()
+ ~CpDecisionManager()
+ onInitialize()
+ getDecisionCounter()
+ nextDecision()
+ resetDecisionCounter()
+ setDecisionCounter()

cl_modbus_tcp_relay
::CpModbusConnection
+ onConnectionLost_
+ onConnectionRestored_
+ onConnectionError_
- ip_address_
- port_
- slave_id_
- heartbeat_interval_ms_
- connect_on_init_
- ctx_
- connected_
- mutex_
- postConnectionLostEvent_
- postConnectionRestoredEvent_
+ CpModbusConnection()
+ ~CpModbusConnection()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ connect()
+ disconnect()
+ reconnect()
+ isConnected()
+ getContext()
+ getMutex()
+ getIpAddress()
+ getPort()
+ getSlaveId()
+ onConnectionLost()
+ onConnectionRestored()
+ onConnectionError()
# update()
- declareAndLoadParam()

cl_modbus_tcp_relay
::CpModbusRelay
+ onWriteSuccess_
+ onWriteFailure_
+ onReadComplete_
+ NUM_CHANNELS
+ COIL_BASE_ADDRESS
- connectionComponent_
- postWriteSuccessEvent_
- postWriteFailureEvent_
+ CpModbusRelay()
+ ~CpModbusRelay()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ writeCoil()
+ readCoil()
+ writeAllCoils()
+ writeAllCoils()
+ readAllCoils()
+ onWriteSuccess()
+ onWriteFailure()
+ onReadComplete()
- channelToAddress()
- isValidChannel()

cl_moveit2z::CpGraspingComponent
+ fingerTipNames
+ gripperLink_
+ currentAttachedObjectName
- graspingObjects
+ getGraspingObject()
+ createGraspableBox()

cl_moveit2z::CpJointSpace
TrajectoryPlanner
- moveit2zClient_
- ikClient_
+ CpJointSpaceTrajectoryPlanner()
+ ~CpJointSpaceTrajectoryPlanner()
+ onInitialize()
+ planFromWaypoints()

cl_moveit2z::CpMotionPlanner
- moveit2zClient_
+ CpMotionPlanner()
+ ~CpMotionPlanner()
+ onInitialize()
+ planToPose()
+ planToJointTarget()
+ planCartesianPath()
+ getCurrentState()
- applyPlanningOptions()

cl_moveit2z::CpTfListener
- tfBuffer_
- tfListener_
- tfMutex_
+ CpTfListener()
+ ~CpTfListener()
+ onInitialize()
+ lookupTransform()
+ transformPose()
+ canTransform()
+ getBuffer()

cl_moveit2z::CpTrajectory
Executor
- moveit2zClient_
- trajectoryHistory_
+ CpTrajectoryExecutor()
+ ~CpTrajectoryExecutor()
+ onInitialize()
+ execute()
+ executePlan()
+ cancel()
+ getTrajectoryHistory()
- applyExecutionOptions()
- recordToHistory()

cl_moveit2z::CpTrajectory
History
- trajectoryHistory_
+ getLastTrajectory()
+ getLastTrajectory()
+ pushTrajectory()

cl_moveit2z::CpTrajectory
Visualizer
- markersPub_
- markers_
- markersMutex_
- markersReady_
- enabled_
- markerScale_
- markerColor_
- publishRate_
- lastPublishTime_
+ CpTrajectoryVisualizer()
+ ~CpTrajectoryVisualizer()
+ onInitialize()
+ update()
+ setTrajectory()
+ clearMarkers()
+ setColor()
+ setScale()
+ setEnabled()
+ isEnabled()

cl_nav2z::CpAmcl
- initalPosePub_
+ CpAmcl()
+ ~CpAmcl()
+ getName()
+ onInitialize()
+ setInitialPose()

cl_nav2z::CpCostmapSwitch
+ layerNames
- costmapProxies
- nav2zClient_
+ CpCostmapSwitch()
+ onInitialize()
+ exists()
+ enable()
+ enable()
+ disable()
+ disable()
+ registerProxyFromDynamic
ReconfigureServer()
+ getStandardCostmapName()

cl_nav2z::CpGoalCheckerSwitcher
- goal_checker_selector
_topic_
- default_goal_checker
_name_

- goal_checker_selector_pub_
+ CpGoalCheckerSwitcher()
+ onInitialize()
+ ~CpGoalCheckerSwitcher()
+ setDefaultGoalChecker()
+ setGoalCheckerId()

cl_nav2z::CpPlannerSwitcher
- desired_planner_
- desired_controller_
- set_planners_mode_flag_
- planner_selector_pub_
- controller_selector_pub_
- goal_checker_selector_pub_
+ CpPlannerSwitcher()
+ onInitialize()
+ setDesiredGlobalPlanner()
+ setDesiredController()
+ commitPublish()
+ setBackwardPlanner()
+ setUndoPathBackwardPlanner()
+ setForwardPlanner()
+ setPureSpinningPlanner()
+ setDefaultPlanners()

cl_nav2z::CpPose
+ isInitialized
+ frozenReferenceFrameTime
- pose_
- poseFrameName_
- referenceFrame_
- m_mutex_
- tfBuffer_
- tfListener_
- listenerMutex_
+ CpPose()
+ CpPose()
+ onInitialize()
+ update()
+ waitTransformUpdate()
+ toPoseMsg()
+ toPoseStampedMsg()
+ getYaw()
+ getX()
+ getY()
+ getZ()
+ setReferenceFrame()
+ getReferenceFrame()
+ getFrameId()
+ freezeReferenceFrame()
+ unfreezeReferenceFrame()

cl_nav2z::CpSlamToolbox
- state_
+ CpSlamToolbox()
+ ~CpSlamToolbox()
+ getState()
+ toggleState()

cl_nav2z::CpWaypointNavigatorBase
+ waypointsEventDispatcher
+ currentWaypoint_
# waypoints_
# waypointsNames_
+ CpWaypointNavigatorBase()
+ ~CpWaypointNavigatorBase()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ loadWayPointsFromFile()
+ loadWayPointsFromFile2()
+ setWaypoints()
+ setWaypoints()
+ getWaypoints()
+ getWaypointNames()
+ getNamedPose()
+ getPose()
+ getCurrentPose()
+ getCurrentWaypointIndex()
+ getCurrentWaypointName()
+ rewind()
+ forward()
+ seekName()
+ loadWaypointsFromYamlParameter()
+ notifyGoalReached()
# insertWaypoint()
# removeWaypoint()

cl_nav2z::CpWaypointsVisualizer
+ waypointsNavigator_
- m_mutex_
- markersPub_
- markers_
- markerLabels_
+ CpWaypointsVisualizer()
+ onInitialize()
# update()
- createMarker()
- createMarkerLabel()

cl_nav2z::components
::CpNav2ActionInterface
+ onNavigationSucceeded_
+ onNavigationAborted_
+ onNavigationCancelled_
+ onNavigationFeedback_
+ postNavigationSuccessEvent
+ postNavigationAbortedEvent
+ postNavigationCancelledEvent
+ postNavigationFeedbackEvent
- actionClient_
+ CpNav2ActionInterface()
+ ~CpNav2ActionInterface()
+ sendNavigationGoal()
+ sendGoal()
+ cancelNavigation()
+ isNavigationServerReady()
+ waitForNavigationServer()
+ onComponentInitialization()
+ onInitialize()
+ getActionClient()
+ onNavigationSucceeded()
+ onNavigationAborted()
+ onNavigationCancelled()
+ onNavigationFeedback()
- onNavigationSuccessCallback()
- onNavigationAbortedCallback()
- onNavigationCancelledCallback()
- onNavigationFeedbackCallback()

cl_nav2z::odom_tracker
::CpOdomTracker
# robotBasePathPub_
# robotBasePathStackedPub_
# odomSub_
# robotPose_
# robotPoseTimer_
# recordPointDistanceThreshold_
# recordAngularDistanceThreshold_
# clearPointDistanceThreshold_
# clearAngularDistanceThreshold_
# odomFrame_
# odomTopicName_
# publishMessages
# baseTrajectory_
# workingMode_
# pathStack_
# pathInfos_
# aggregatedStackPathMsg_
# strategy_
# currentMotionGoal_
# currentPathName_
# param_callback_handle_
# m_mutex_
+ CpOdomTracker()
+ processNewPose()
+ odomMessageCallback()
+ update()
+ setWorkingMode()
+ setPublishMessages()
+ pushPath()
+ pushPath()
+ popPath()
+ clearPath()
+ setStartPoint()
+ setStartPoint()
+ setCurrentMotionGoal()
+ setCurrentPathName()
+ getCurrentMotionGoal()
+ getPath()
+ logStateString()
+ updateParameters()
+ setOdomFrame()
# onInitialize()
# updateConfiguration()
# rtPublishPaths()
# updateRecordPath()
# updateClearPath()
# updateAggregatedStackPath()
# parametersCallback()

cl_px4_mr::CpGoalChecker
+ onGoalReached_
- localPosition_
- goalX_
- goalY_
- goalZ_
- xyTolerance_
- zTolerance_
- goalActive_
+ CpGoalChecker()
+ ~CpGoalChecker()
+ onInitialize()
+ update()
+ setGoal()
+ clearGoal()
+ isGoalActive()

cl_px4_mr::CpMicroRosAgent
- agentPid_
- launched_
- shutdownRequested_
- command_
- nodeName_
- mutex_
- pipeFd_
+ CpMicroRosAgent()
+ ~CpMicroRosAgent()
+ launch()
+ shutdown()
+ isLaunched()
+ getPid()
+ getNodeName()
- killProcessesRecursive()

cl_px4_mr::CpOffboardKeep
Alive
- publisher_
- trajectorySetpoint_
- enabled_
+ CpOffboardKeepAlive()
+ ~CpOffboardKeepAlive()
+ onInitialize()
+ update()
+ enable()
+ disable()
+ isEnabled()

cl_px4_mr::CpTrajectorySetpoint
- publisher_
- localPosition_
- lastSetpoint_
- mutex_
- hasPublished_
+ CpTrajectorySetpoint()
+ ~CpTrajectorySetpoint()
+ onInitialize()
+ setPositionNED()
+ hold()
+ republishLast()
+ getLastSetpoint()

cl_px4_mr::CpVehicleCommand
- publisher_
+ CpVehicleCommand()
+ ~CpVehicleCommand()
+ onInitialize()
+ sendCommand()
+ arm()
+ forceArm()
+ disarm()
+ setOffboardMode()
+ land()
+ takeoff()

cl_px4_mr::CpVehicleCommandAck
+ onAckReceived_
- subscriber_
- lastCommand_
- lastResult_
+ CpVehicleCommandAck()
+ ~CpVehicleCommandAck()
+ onInitialize()
+ getLastAckCommand()
+ getLastAckResult()
- onAckMessage()

cl_px4_mr::CpVehicleLocal
Position
+ onPositionReceived_
- subscriber_
- x_
- y_
- z_

- heading_
- valid_
- mutex_
+ CpVehicleLocalPosition()
+ ~CpVehicleLocalPosition()
+ onInitialize()
+ getX()
+ getY()
+ getZ()
+ getHeading()
+ isValid()
- onPositionMessage()

cl_px4_mr::CpVehicleStatus
+ onArmed_
+ onDisarmed_
+ onModeChanged_
+ onLanded_
- subscriber_
- armingState_
- navState_
- prevArmingState_
- prevNavState_
- mutex_
+ CpVehicleStatus()
+ ~CpVehicleStatus()
+ onInitialize()
+ isArmed()
+ isLanded()
+ getNavState()
+ getArmingState()
- onStatusMessage()

smacc2::client_core
_components::CpActionClient
< ActionType >
+ actionServerName
+ serverTimeout
+ onActionSucceeded_
+ onActionAborted_
+ onActionCancelled_
+ onActionFeedback_
+ onGoalAccepted_
+ onGoalRejected_
+ postSuccessEvent
+ postAbortedEvent
+ postCancelledEvent
+ postFeedbackEvent
- client_
- lastRequest_
- lastCancelResponse_
- feedbackCallback_
- actionMutex_
+ CpActionClient()
+ CpActionClient()
+ ~CpActionClient()
+ sendGoal()
+ cancelGoal()
+ isServerReady()
+ waitForServer()
+ onInitialize()
+ onComponentInitialization()
+ onSucceeded()
+ onAborted()
+ onCancelled()
+ onFeedback()
+ getActionClient()
- onFeedback()
- onResult()
- postResultEvent()

smacc2::client_core
_components::CpRos2Timer
+ onTimerTick_
- timer_
- duration_
- oneshot_
- initialized_
+ CpRos2Timer()
+ ~CpRos2Timer()
+ onInitialize()
+ startTimer()
+ stopTimer()
+ cancelTimer()
+ onTimerTick()
- timerCallback()

smacc2::client_core
_components::CpServiceClient
< ServiceType >
+ serviceName
+ serviceTimeout
+ onServiceResponse_
+ onServiceRequestSent_
+ onServiceFailure_
+ postResponseEvent
+ postRequestSentEvent
+ postFailureEvent
- client_
- lastRequest_
- serviceMutex_
+ CpServiceClient()
+ CpServiceClient()
+ CpServiceClient()
+ ~CpServiceClient()
+ sendRequest()
+ sendRequestSync()
+ isServiceReady()
+ waitForService()
+ onInitialize()
+ onComponentInitialization()
+ onResponse()
+ onRequestSent()
+ onFailure()
+ getServiceClient()

smacc2::client_core
_components::CpSubprocessExecutor
+ onCommandCompleted_
+ onCommandFailed_
- initialized_
- execution_mutex_
+ CpSubprocessExecutor()
+ ~CpSubprocessExecutor()
+ onInitialize()
+ executeCommand()
+ executeCommandAsync()
+ onCommandCompleted()
+ onCommandFailed()

smacc2::client_core
_components::CpTopicPublisher
< MessageType >
+ queueSize
+ durability
+ reliability
- pub_
- initialized_
- topicName_
+ CpTopicPublisher()
+ ~CpTopicPublisher()
+ publish()
+ onInitialize()

smacc2::client_core
_components::CpTopicSubscriber
< MessageType >
+ queueSize
+ postMessageEvent
+ postInitialMessageEvent
+ onFirstMessageReceived_
+ onMessageReceived_
- sub_
- firstMessage_
- initialized_
- topicName_
+ CpTopicSubscriber()
+ CpTopicSubscriber()
+ ~CpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ onInitialize()
- messageCallback()

< TMessageType >

cl_nav2z::CpWaypointNavigator
+ client_
+ nav2ActionInterface_
+ onNavigationRequestSucceded
+ onNavigationRequestAborted
+ onNavigationRequestCancelled
- succeddedNav2ZClientConnection_
- abortedNav2ZClientConnection_
- cancelledNav2ZClientConnection_
+ CpWaypointNavigator()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ sendNextGoal()
+ stopWaitingResult()
- onNavigationResult()
- onGoalReached()
- onGoalCancelled()
- onGoalAborted()

< nav2_msgs::action
::NavigateToPose > < ActionT > < vision_msgs::msg

::Detection3DArray > < TMessageType > < AprilTagDetectionArray >